Monday

MEMO

M-Number: [ ][ ][ ][ ]

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WTW158 Calculus

Class Test 4 [Monday] 22–26 April 2024 Time: 15 min

Surname: Initials:

Student number: Group:

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2 marks] Let $f(x) = \dfrac{5x^2 - 2x + 7}{3x^2 - 1}$. Find the horizontal asymptote of f, if any.

$$\lim_{x\to\infty} \frac{5x^2-2x+7}{3x^2-1} \quad \left(\frac{\infty}{\infty}\right)$$

You must take both limits!!

$$= \lim_{x\to\infty} \frac{(5x^2-2x+7)^{1/x^2}}{(3x^2-1)^{1/x^2}}$$

$$= \lim_{x\to\infty} \frac{5-2/x+7/x^2}{3-1/x^2} = 5/3$$

$$\lim_{x\to-\infty} \frac{5x^2-2x+7}{3x^2-1} \left(\frac{\infty}{\infty}\right) = \lim_{x\to-\infty} \frac{5-2/x+7/x^2}{3-1/x^2} = 5/3$$

$\Rightarrow$ $y = 5/3$ is the horizontal asymptote !!

QUESTION 2 [1 marks] Write dawn the definition of a derivative of a function f at a number $x = a$.

$$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$$

OR: $f'(a) = \lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h}$.

QUESTION 3 [2, 2 marks] Consider the function $f(x) = \sqrt{x+1}$.

1. Use the definition of the derivative to show that $f'(x) = \dfrac{1}{2\sqrt{x+1}}$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \left(\frac{\sqrt{x+h+1} - \sqrt{x+1}}{h}\right)\left(\frac{\sqrt{x+h+1} + \sqrt{x+1}}{\sqrt{x+h+1} + \sqrt{x+1}}\right)$$

$$= \lim_{h\to 0} \frac{x+h+1-(x+1)}{h\left[\sqrt{x+h+1} + \sqrt{x+1}\right]} = \lim_{h\to 0} \frac{h}{h\left(\sqrt{x+h+1} + \sqrt{x+1}\right)}$$

$$= \frac{1}{2\sqrt{x+1}}$$

2. Find the equation of the line tangent to the graph of $f(x)$ at $x = 1$.

$f(1) = \sqrt{2}$

$f'(1) = \dfrac{1}{2\sqrt{2}}$ (from 1)

$$y - f(1) = f'(1)(x-1)$$

$$y - \sqrt{2} = \frac{1}{2\sqrt{2}}(x-1)$$

You can leave the answer in this form.

$$y = \frac{1}{2\sqrt{2}}x - \frac{1}{2\sqrt{2}} + \sqrt{2}.$$

QUESTION 4 [1 mark] Find $g'(x)$, given $g(x) = x^5e^x$.

$$g(x) = x^5e^x$$

$$g'(x) = 5x^4e^x + x^5e^x.$$

NB!

You $\boxed{\text{may}}$ have: $g'(x) = \frac{d}{dx}(x^5)e^x + x^5\frac{d}{dx}(e^x)$

$$= 5x^4e^x + x^5e^x$$

Please note that the notation here is important.

Tuesday.

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QUESTION 1 [2 marks] Find the horizontal and vertical asymptotes of $f(x) = \dfrac{x+1}{(x+1)(x-3)}$, if any.

Horizontal asymptotes:

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{\cancel{x+1}}{\cancel{(x+1)}(x-3)} = \lim_{x\to\infty} \frac{1}{x-3} = 0$$

$$\lim_{x\to-\infty} f(x) = \lim_{x\to-\infty} \frac{1}{x-3} = 0$$

$y=0$ is the horizontal asymptote of f.

$$f(x) = \frac{x+1}{(x+1)(x-3)} = \frac{1}{x-3}, \quad x \neq -1.$$

Vertical asymptote:

$$\lim_{x\to3^+} f(x) = \lim_{x\to3^+} \frac{1}{x-3} = +\infty$$

$x \to 3^+$
$\Rightarrow x > 3$
$\Rightarrow x-3 > 0$

$\Rightarrow x=3$ is a vertical asymptote of f.

$$f(x+h) = \sqrt{1-(x+h)} = \sqrt{1-x-h}$$

QUESTION 2 [2, 2 marks] Consider the function $f(x) = \sqrt{1-x}$.

1. Use the definition to show that $f'(x) = \dfrac{-1}{2\sqrt{1-x}}$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \left(\frac{\sqrt{1-x-h} - \sqrt{1-x}}{h} \right) \left(\frac{\sqrt{1-x-h} + \sqrt{1-x}}{\sqrt{1-x-h} + \sqrt{1-x}} \right)$$

$$= \lim_{h \to 0} \frac{(1-x-h) - (1-x)}{h\left[\sqrt{1-x-h} + \sqrt{1-x}\right]} \qquad 1-x-h-1+x = -h$$

$$= \lim_{h \to 0} \frac{-h}{h(\sqrt{1-x-h} + \sqrt{1-x})}$$

$$= \lim_{h \to 0} \frac{-1}{\sqrt{1-x-h} + \sqrt{1-x}} = \frac{-1}{2\sqrt{1-x}} =$$

2. Find the equation of the line tangent to the graph of $f(x)$ at $x = -1$.

$$y - f(-1) = f'(-1)(x+1) \qquad f(-1) = \sqrt{1+1} = \sqrt{2}$$

$$y - f(-1) = \frac{-1}{2\sqrt{2}}(x+1) \qquad f'(-1) = \frac{-1}{2\sqrt{2}}$$

$\Rightarrow$ $y - \sqrt{2} = \dfrac{-1}{2\sqrt{2}}(x+1)$ is the equation of the tangent at $x=-1$.

QUESTION 3 [2 marks] Let $g(x) = x^2 + 3x$ and $f(x) = 5e^x$. Find the derivative of the quotient function $\dfrac{f(x)}{g(x)}$.

$$\frac{d}{dx}\left(\frac{5e^x}{x^2+3x} \right)$$

NB: Notation!!

$$= \frac{\frac{d}{dx}(5e^x)(x^2+3x) - 5e^x \frac{d}{dx}(x^2+3x)}{(x^2+3x)^2}$$

$$= \frac{5e^x(x^2+3x) - 5e^x(2x+3)}{(x^2+3x)^2}.$$

This step is not necessary.

DO NOT SIMPLIFY!!

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QUESTION 1 [3 marks]

Show that $\frac{d}{dx}(f(x)g(x)) = \frac{d}{dx}(f(x))g(x) + f(x)\frac{d}{dx}g(x)$ by using the definition of a derivative.

$$\frac{d}{dx} f(x)g(x) = \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x+h)g(x)}{h} + \lim_{h\to 0} \frac{f(x+h)g(x) - f(x)g(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h)(g(x+h) - g(x))}{h} + \lim_{h\to 0} \frac{(f(x+h) - f(x))g(x)}{h}$$

$$= \lim_{h\to 0} f(x+h) \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} + \left[\lim_{h\to 0} \frac{f(x+h) - f(x)}{h}\right] \times g(x)$$

$$= f(x)g'(x) + f'(x)g(x)$$

$$= f'(x)g(x) + f(x)g'(x).$$

[2, 1]

QUESTION 2 ~~[$1\frac{1}{2}, \frac{1}{2}$~~ marks] Consider the function $f(x) = \frac{\cos x}{e^{2x}}$.

1. Evaluate $\lim_{x\to\infty} f(x)$.

$$-1 \le \cos x \le 1$$

$$\Rightarrow \frac{-1}{e^{2x}} \le \frac{\cos x}{e^{2x}} \le \frac{1}{e^{2x}}$$

$$\lim_{x\to\infty} \frac{-1}{e^{2x}} = 0$$

$$\lim_{x\to\infty} \frac{1}{e^{2x}} = 0$$

$\Rightarrow$ By the squeeze theorem, $\lim_{x\to\infty} \frac{\cos x}{e^{2x}} = 0$

2. Find the horizontal asymptote(s) of $f(x)$, if they exist.

$$\lim_{x\to\infty} \frac{\cos x}{e^{2x}} = 0$$

$$\lim_{x\to-\infty} \frac{\cos x}{e^{2x}} = \lim_{x\to-\infty} \cos x\, e^{-2x} = \infty$$

$y=0$ is a horizontal asymptote of f.

QUESTION 3 [2 marks] Find $g'(x)$, given $g(x) = \frac{3e^x}{e^x + 2x}$.

$$\frac{d}{dx}\left(\frac{3e^x}{e^x+2x}\right)$$

$$= \frac{3e^x(e^x+2x) - 3e^x(e^x+2)}{(e^x+2x)^2}$$

$$-1 \le \cos x \le 1$$

$$\frac{-1}{e^{+2x}} \le \frac{\cos x}{e^{2x}} \le \frac{1}{e^{2x}}$$

$$\lim_{x\to-\infty} \frac{-1}{e^{2x}} = \lim_{x\to-\infty} -e^{-2x} = \infty, \text{ also } \lim_{x\to-\infty} \frac{1}{e^{2x}} = \lim_{x\to-\infty} e^{-2x} = \infty$$

Memo - students.

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Surname: ______________________ Initials: ______

Student number: ______________ Group: ______

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2, 2 marks] Consider the function $f(x) = \sqrt{2x-1}$.

1. Use the definition of the derivative to find $f'(x)$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sqrt{2(x+h)-1} - \sqrt{2x-1}}{h}$$

$$= \lim_{h\to 0} \frac{\sqrt{2x+2h-1} - \sqrt{2x-1}}{h}\left(\frac{\sqrt{2x+2h-1} + \sqrt{2x-1}}{\sqrt{2x+2h-1} + \sqrt{2x-1}}\right)$$

$$= \lim_{h\to 0} \frac{(2x+2h-1) - (2x-1)}{h(\sqrt{2x+2h-1} + \sqrt{2x-1})}$$

$$= \lim_{h\to 0} \frac{2h}{h(\sqrt{2x+2h-1} + \sqrt{2x+1})} = \frac{2}{2\sqrt{2x-1}}$$

$$= \frac{1}{\sqrt{2x-1}}$$

2. Find the equation of the line tangent to the graph of $f(x)$ at the point $(1, 1)$.

$$f'(1) = \frac{1}{\sqrt{2(1)-1}} = \frac{1}{\sqrt{2-1}} = 1$$

$$y - f(1) = m(x-1)$$

$$= f'(1)(x-1)$$

$$\Rightarrow y - 1 = 1(x-1)$$

$$\Rightarrow \underline{y = x}.$$

[2, 1]

QUESTION 2 [~~1½~~, ~~½~~ marks]

Let $f(x) = \frac{\sin x}{x^4+1}$.

1. Evaluate $\lim_{x\to\infty} f(x)$.

$$-1 \le \sin x \le 1$$

$$\frac{-1}{x^4+1} \le \frac{\sin x}{x^4+1} \le \frac{1}{x^4+1}$$

$$\lim_{x\to\infty} \frac{-1}{x^4+1} = 0$$

$$\lim_{x\to\infty} \frac{1}{x^4+1} = 0$$

$\Rightarrow$ By the squeeze theorem, $\lim_{x\to\infty} \frac{\sin x}{x^4+1} = 0$

2. Find the horizontal asymptote(s) of $f(x)$, if they exist.

From (1): $y=0$ is a horizontal asymptote

$\lim_{x\to-\infty} \frac{\sin x}{x^4+1}$ must be obtained

Also consider this limit.

$$-1 \le \sin x \le 1$$

$$\frac{-1}{x^4+1} \le \frac{\sin x}{x^4+1} \le \frac{1}{x^4+1}$$

$\Rightarrow$ By the squeeze theorem, since $\lim_{x\to-\infty} \frac{-1}{x^4+1} = 0$ and $\lim_{x\to\infty} \frac{1}{x^4+1} = 0$, we have: $\lim_{x\to-\infty} f(x) = 0$

$\Rightarrow$ only $y=0$ is a Horizontal asymptote.

QUESTION 3 [2 marks] Find $g'(x)$, given $g(x) = \frac{2x^8-1}{3e^x}$.

$$\frac{d}{dx}\left(\frac{2x^8-1}{3e^x}\right)$$

$$= \frac{(16x^7)(3e^x) - (2x^8-1)3e^x}{(3e^x)^2}$$